## 44 Putting lives in context: geospatial reconstructions of landcover from pollen records using the Multiple Scenario Approach

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Human activity, past and present, takes place in a spatial landscape which both constrains human activity and is shaped by it. Pollen records are point data which preserve a complex distance-weighted record of the surrounding vegetation, a key component in determining past activities and resource availability. This paper explores the Multiple Scenario Approach (MSA), which uses geospatial computing to reconstruct estimates of abundance and arrangement of vegetation in past landscapes, quantifying the context for human activity and supporting the development and testing of archaeologically-based hypotheses for human-environment interactions.

The MSA is a GIS-first approach to reconstruction, which creates maps simulating various possible landcover arrangements, then models the pollen signal that would be produced and compares it with actual pollen records to identify which scenarios are possible.

Originally developed in the early 2000s, MSA software was intended to be accessible to the whole environmental archaeology community and therefore was based on freely available data formats and assumptions about future systems and software licenses. The last 25 years have seen many changes in computer systems, reducing the accessibility and shareability of the original package. In 2020 we began to develop a new software package for the MSA approach, MSA-Q, in QGIS and Python using FAIR principles, and through that process have had the opportunity to reflect on our own assumptions. Initial applications illustrated the challenges and benefits of communicating the limitations and uncertainties associated with reconstruction to collaborators, and of developing effective workflows to integrate pollen-based reconstruction with other evidence.

This paper will outline lessons learned and future challenges to address in creating open source software for geospatial landcover reconstruction of dynamic cultural landscapes.